Physics Worksheet: Propagation & Reflection of Light

Name: ______________________________ **Date:** _______________ **Grade 10**

Part 1: Multiple Choice

Circle the correct answer for each of the following questions.

1. According to the principle of rectilinear propagation, how does light travel in a homogeneous transparent medium?

a) In zigzag patterns
b) In straight lines
c) In circular waves
d) It bends randomly

2. What do we call a beam of light where all the rays come together to meet at a single point?

a) Converging beam
b) Diverging beam
c) Parallel (Cylindrical) beam
d) Scattered beam

3. We are able to see non-luminous objects (like a desk or a book) because:

a) They emit their own light
b) They refract light passing through them
c) They reflect light towards our eyes
d) They are transparent

4. You stand in front of a flat, plane mirror. Which statement best describes the image formed?

a) Real and larger than you
b) Real and smaller than you
c) Virtual and smaller than you
d) Virtual and of the same size

5. Which of the following is considered a Luminous Object (a primary source of light)?

a) The Moon
b) A Plane Mirror
c) The Sun

d) A White Piece of Paper

6. When parallel light rays strike a rough, uneven surface and bounce off in many different directions, this is known as:

a) Specular (Regular) reflection
b) Diffuse reflection
c) Refraction
d) Absorption

Part 2: Conceptual Questions & Calculations

7. **The Laws of Reflection:** When a laser strikes a flat mirror, it bounces off. State the *two laws of reflection* that govern this behavior.

8. **Types of Reflection:** You observe a puddle on the street. When the water is perfectly calm, you see a clear reflection of a streetlamp. When it starts to rain, the water ripples and the reflection becomes distorted.
Explain this ”Puddle Effect” using the physics concepts of **Regular** and **Diffuse** reflection.

9. **Spectrum & Color:** A green leaf is illuminated by white light (which contains all colors of the visible electromagnetic spectrum). Explain why the human eye perceives the leaf as green.

10. **Reflection Calculation:** Suppose you are holding a flat mirror and standing at the exact center of a giant clock face built into the floor. A friend standing at the 12 o'clock mark shines a beam of light toward you. You want to use the mirror to reflect the beam exactly toward another observer standing at the 5 o'clock mark.
(Hint: A full circle is $360°$ *and has 12 hours. Calculate the angle between 12 o'clock and 5 o'clock first.)*

a) What is the total angle between the incident ray and the reflected ray?

b) What must the **angle of incidence ($\hat{i}$)** be to achieve this?

c) What must the **angle of reflection ($\hat{r}$)** be?

Part 3: Ray Diagrams & Plane Mirrors

11. **Drawing Reflection:** In the space below, draw a ray diagram showing a single ray of light striking a plane mirror. You must clearly label the following:
    - The incident ray
    - The reflected ray
    - The normal line
    - The angle of incidence (i)
    - The angle of reflection (r)

Student Drawing Area
Use a ruler to construct your ray diagram.

12. **Image Characteristics:** Suppose you place a pencil in front of a flat plane mirror. List **four** characteristics of the image that is formed by the mirror.

End of Worksheet